

Password Strength Analyzer – Cyber Security Project

This project is a Cyber Security tool that checks whether a password is weak, medium, or strong. It helps users understand password security and create safer passwords.

1. Project objective

To develop a password strength analysers that evaluates a user's password based on length, complexity, and common password patterns, and provides suggestions to improve password security.

2. Key features

Password length checking

Checks whether the password has enough characters. Longer passwords are generally harder to guess.

Complexity checking

Checks for uppercase letters, lowercase letters, numbers, and special characters.

Common password detection

Identifies easily guessed passwords, such as 123456, password, or repeated characters.

Password improvement suggestions

Provides tips for creating longer, more difficult-to-guess passwords.

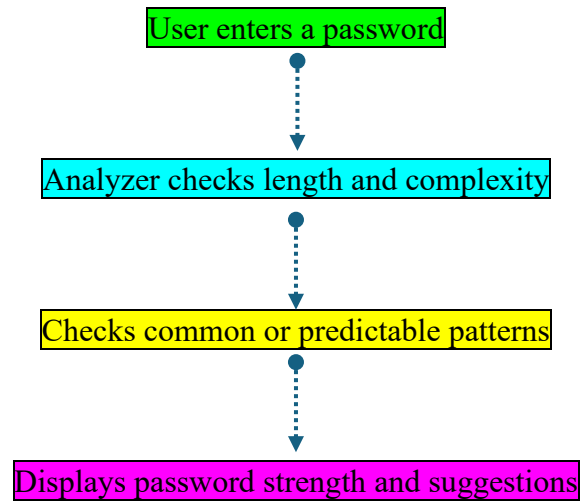
Password reuse prevention

Can check whether a password has been used before, using securely stored password hashes rather than plain-text passwords.

3. Technologies used

- Python – Main programming language.
- Regular Expressions (Regex) – To check password patterns.
- Tkinter – To create a simple graphical user interface.
- Hashing (optional) – To demonstrate secure password storage concepts.

4. How the project works



5. Python project implementation

Here is a beginner-friendly password strength analyzer using Python and Tkinter.

```
import tkinter as tk
from tkinter import messagebox
import re

def analyze_password():
    password = password_entry.get()

    if not password:
        messagebox.showwarning("Warning", "Please enter a password")
        return

    score = 0
    suggestions = []

    # Check password length
    if len(password) >= 12:
        score += 2
    elif len(password) >= 8:
        score += 1
    else:
        suggestions.append("Use at least 12 characters")

    # Check uppercase letters
    if re.search(r"[A-Z]", password):
        score += 1
    else:
        suggestions.append("Add uppercase letters")

    # Check lowercase letters
    if re.search(r"[a-z]", password):
        score += 1
    else:
        suggestions.append("Add lowercase letters")
```

```

# Check numbers
if re.search(r"[0-9]", password):
    score += 1
else:
    suggestions.append("Add numbers")

# Check special characters
if re.search(r"^[A-Za-z0-9]", password):
    score += 1
else:
    suggestions.append("Add special characters")

# Check common passwords
common_passwords = [
    "password", "123456", "12345678",
    "qwerty", "admin"
]

if password.lower() in common_passwords:
    score = 0
    suggestions.append("Avoid common passwords")

# Display strength
if score <= 2:
    strength = "Weak"
elif score <= 4:
    strength = "Medium"
else:
    strength = "Strong"

result_label.config(
    text=f"Password Strength: {strength}"
)

if suggestions:
    suggestion_label.config(
        text="Suggestions:\n• " + "\n• ".join(suggestions)
    )
else:
    suggestion_label.config(
        text="Good password characteristics!"
    )

# Create window
window = tk.Tk()
window.title("Password Strength Analyzer")
window.geometry("450x350")

title_label = tk.Label(
    window,
    text="Password Strength Analyzer",
    font=("Arial", 18, "bold")
)
title_label.pack(pady=20)

```

```
password_entry = tk.Entry(
window,
width=35,
show="*")
password_entry.pack(pady=10)

check_button = tk.Button(
window,
text="Check Password",
command=analyze_password
)

check_button.pack(pady=10)

result_label = tk.Label(
window,
text="Password Strength: ",
font=("Arial", 14)
)
result_label.pack(pady=10)

suggestion_label = tk.Label(
window,
text="",
justify="left",
wraplength=400
)
suggestion_label.pack(pady=10)

window.mainloop()
```

6. How to run the project

- 1.Open VS Code and create a file named `password_analyzer.py`.
- 2.Copy the Python code into the file.
- 3.Run the program:

```
python password_analyzer.py
```

- 4.Enter a password and click Check Password.

7. Expected output

Password Strength Analyzer

Sample output

Password entered

Hello@123

Password Strength

Medium

Suggestions

- Use at least 12 characters.

Another example

MySecure@2026!

Password Strength

Strong

These are illustrative results from the simple scoring rules in the code, not a guarantee that a password is secure.

8. Conclusion

The Password Strength Analyzer helps users understand password security by checking length, complexity, and common patterns. The project introduces basic cybersecurity concepts such as password protection, pattern matching, and secure password handling.

Important security note: This is an educational analysers. Real password systems should use secure password hashing, such as Argon2id, and should never store or log users' plain-text passwords.